

Aditya Kuchhal

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production-ready ML systems and Generative AI applications. Specialized in Large Language Models, transformer architectures, and cloud-native ML deployment. Deployed ML system processing 6,800+ records with 76.3% accuracy and 16.5ms latency. Published researcher in computer vision with strong expertise in Python, TensorFlow, PyTorch, LangChain, and AWS. Passionate about building intelligent systems that solve real-world business problems at scale.

SKILLS

AI/ML: Large Language Models (GPT, Claude), Generative AI, Deep Learning, Computer Vision, NLP, Transformers, RAG Systems, Prompt Engineering

Frameworks: TensorFlow, PyTorch, scikit-learn, Keras, Hugging Face Transformers, LangChain, LlamaIndex, YOLOv8

Languages: Python (NumPy, pandas, Matplotlib, OpenCV), JavaScript, Java, SQL

Databases: PostgreSQL, MySQL, MongoDB, Redis, Vector Databases (Pinecone, Chroma, FAISS)

Cloud & MLOps: AWS (SageMaker, EC2, Lambda, S3), GCP, Docker, CI/CD, GitHub Actions, Model Deployment

Data Visualization: Power BI (Power Apps), Tableau, Matplotlib, Seaborn, Snowflake, Alteryx, ETL Pipelines

Tools & Methodologies: Git, Agile/Scrum, JIRA, Jupyter, FastAPI, Flask, REST APIs, TDD, Design Patterns, Microservices

EDUCATION

George Brown College - Toronto, ON | *Post-Graduate Diploma*

- Applied A.I. Solutions Development, August 2025
- Coursework: Machine Learning, Deep Learning, Python, Neural Networks, NLP, Computer Vision, MLOps

York University - Toronto, ON | *Post-Graduate Diploma*

- Back-End and Blockchain Development, May 2024
- Coursework: Backend Architecture, Microservices, RESTful APIs, DevOps, CI/CD

Amity University - Noida, India | *Bachelor of Technology*

- Computer Science and Engineering, July 2023
 - Majors: Data Structure & Algorithms, Database Management System, Object-Oriented Programming
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WORK EXPERIENCE

AI/ML Engineer | May 2025 to August 2025 | **Kinectrics**

- Designed and deployed **production ML system** processing **6,800+ records** using **Python, TensorFlow, and scikit-learn**, achieving **76.3% accuracy** with optimized neural network architecture
- Built **end-to-end ML pipeline** including data preprocessing, feature engineering, model training, and inference API, reducing per-record latency to **16.5ms**
- Implemented **model monitoring infrastructure** tracking performance metrics and data drift in real-time, improving system reliability by 30%
- Collaborated with cross-functional teams using **Agile methodology**, conducting code reviews and technical presentations, improving sprint velocity by 20%

Software Developer | June 2022 to August 2022 | **Genpact**

- Developed **AI automation platform** using **Python and ML models**, reducing operational processing time by **30%** and improving throughput by **35%**

- Built **automated classification system** using scikit-learn and XGBoost for ticket categorization, achieving 85%+ accuracy and lowering resolution time by **40%**
- Designed **REST APIs** for model serving with comprehensive error handling and monitoring
- Implemented **data preprocessing pipelines** using pandas and NumPy for model training and inference

Software Engineer | March 2022 to May 2022 | **AgroCep**

- Deployed **cloud-native ML microservices** on AWS (EC2, Lambda, S3), reducing inference latency by **35%** and infrastructure costs by **25%**
 - Built **CI/CD pipelines** for **automated model deployment** using **Docker and GitHub Actions**, enabling 50% faster deployment cycles
 - Developed **RESTful APIs** for model inference using Flask, serving 1000+ daily requests with 99.9% uptime
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PERSONAL/ACADEMIC PROJECTS

- **RAG-Powered Question Answering System**

Python, LangChain, OpenAI API, Pinecone, FastAPI

Built production-ready RAG system using LangChain, GPT-4, and Pinecone for context-aware question answering. Implemented document ingestion pipeline with embeddings and vector storage. Designed FastAPI inference API achieving sub-2-second latency. Deployed on AWS Lambda handling 500+ queries/day.

- **Multi-Agent AI Workflow System**

Python, LangChain, Claude API, LangGraph, Redis

Architected multi-agent system with specialized agents for data extraction, analysis, and reporting. Implemented agent orchestration with LangGraph enabling autonomous collaboration. Built state management using Redis. Achieved 60% reduction in manual processing time with 95%+ accuracy.

- **Deep Learning Image Classification (Published)**

Python, TensorFlow, CNN, OpenCV

Developed CNN-based model for degraded script classification achieving 85%+ accuracy. Implemented advanced preprocessing pipeline and multi-stage feature extraction. Published in International Journal of Computing and Digital Systems (IJCDs). Built inference API with Flask deployed on AWS.

- **Predictive Analytics Platform**

Python, scikit-learn, XGBoost, PostgreSQL, React

Built end-to-end ML platform with gradient boosting models for workforce analytics. Designed feature engineering pipeline processing 50+ features. Implemented training framework with cross-validation and hyperparameter tuning. Achieved 78% prediction accuracy on employee turnover.

- **NLP Sentiment Analysis**

Python, BERT, Hugging Face Transformers, Flask

Fine-tuned BERT model for sentiment analysis on customer feedback achieving 92% F1-score. Implemented transfer learning pipeline with domain-specific fine-tuning. Deployed inference API with batch processing support.

CERTIFICATIONS

- Google Data Analysis with R Programming – Coursera
- IBM Python for Data Science – Cognitive Class
- Data Warehousing and Business Intelligence – Coursera
- Pattern Discovery in Data Mining – Coursera
- Text Mining and Analytics - Coursera
- The Python Mega Course – Udemy
- Data Structures and Algorithms Essentials using C++ - Udemy